

Mihai Nicola

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RESEARCH INTERESTS

My research aims to advance software reliability. In my work I apply formal methods to develop new techniques for reasoning about program correctness. I focus in particular on programming languages, automated verification, and concurrency.

EDUCATION

Stevens Institute of Technology

Ph.D. in Computer Science

Hoboken, NJ

Sept 2021 - Present

Politehnica University

M.S. in Computer Science

Bucharest, Romania

Sept 2018 - Sept 2020

Faculty of Cybernetics, ASE

B.S. in Economic Informatics

Bucharest, Romania

Sept 2000 - Jun 2005

PUBLICATIONS

Nicola, M., Enea, C., & Koskinen, E (*under submission*). Automating Induction-Reduction for Concurrent Object Proofs.

Nicola, M., Agarwal, C., Koskinen, E., & Wies, T. (2025). Abstract Interpretation of Temporal Safety Effects of Higher Order Programs. *Proceedings of the ACM on Programming Languages*, 9(OOPSLA2), 2511-2539.

Chen, A., Fathololumi, P., **Nicola, M.**, Pincus, J., Brennan, T., & Koskinen, E. (2023, October). Better predicates and heuristics for improved commutativity synthesis. In *International Symposium on Automated Technology for Verification and Analysis* (pp. 93-113). Cham: Springer Nature Switzerland.

AWARDS & HONORS

Provost Doctoral Fellowship. Stevens Institute of Technology (2021)

TALKS

Automated Refinement of Concurrent Object Quotients. VDS 2025

Inferring Dependent Types and Effects through Abstract Interpretation NJPLS May 2024

TEACHING EXPERIENCE

Teaching Assistant

Stevens Institute of Technology

Compiler Design. Spring 2024, Spring 2025

Concurrent Programming. Fall 2025

PROJECTS

Mechanization of Correctness Proofs for Concurrent Objects in $\text{L}\lambda\text{E}\lambda\text{N}$

- Ongoing work exploring the mechanization and automation of safety proofs for lock-free concurrent objects.
- Developing a compositional proof framework that derives trace quotients by combining partial-order reduction and Lipton's reduction via induction-reduction tuples capturing recurrent nonblocking synchronization patterns.
- Demonstrating expressiveness by verifying canonical lock-free data structures, including a concurrent counter, Treiber Stack, Michael-Scott Queue, Elimination Stack.

evDrift: Static Analysis of Dependent Temporal Safety Effects of Functional Programs

- Developed a scalable static analysis for automated verification of temporal safety properties over effect traces for a subset of OCaml, equipped with an `ev` expression that emits events as side effects

- Built an automata-based abstract interpreter for the trace extension operator `ev`, integrating it into the abstract-interpretation-based dataflow refinement type inference tool `Drift`
- Demonstrated improved scalability and performance compared to state-of-the-art automated safety verifiers

Servois2: Synthesis of commutativity conditions of data structure methods and code

- Developed a model-counting-guided heuristic for predicate selection during synthesis, providing quantitative coverage criteria over the search space
- Supported rich logical fragments, including linear integer arithmetic (LIA) over bounded domains, and theories of strings, Booleans and arrays.

ACADEMIC SERVICE

Artifact Evaluation Committee. CAV 2023, CAV 2025

External reviewer. NETYS 2024, POPL 2025, OOPSLA 2025

PROFESSIONAL EXPERIENCE

Applied Scientist Intern

Jun - Aug 2026

AWS

Arlington, VA, USA

Joined the Automated Reasoning Group within the Science of Security team to formally reason about safety of cloud infrastructure

Domain Software Architect

2019 – 2021

ING Bank

Bucharest, Romania

Design and validation of reliability-critical systems, focusing on correctness, and guiding development teams to ensure safety under availability constraints.

Consultant / Senior Software Engineer

2006-2019

United Nations Conference on Trade and Development

Geneva, Switzerland

Led the design and implementation of configurable, large-scale software systems for international trade and e-government platforms, deployed in more than 30 countries

Additional industry positions omitted for brevity and available upon request.